Math 190-1: Mathematics of Sustainability	Spring 2026

Lecture 20: Limit Distributions

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By the end of this lecture, you should be able to

- Know what an expected time of return is,
- know how to calculate it, and
- know the relationships between stationary distributions and expected return time.

20.1 Limit Behaviors

Remark 20.1. *We spent a lot of time talking about the even-odd parity of the Ehrenfest chain on Monday, but I forgot the most important part: We need to show that for this problem (because it is not aperiodic), that* $\lim_{n\to\infty} p^n(x,y) \not\to \pi(y)$.

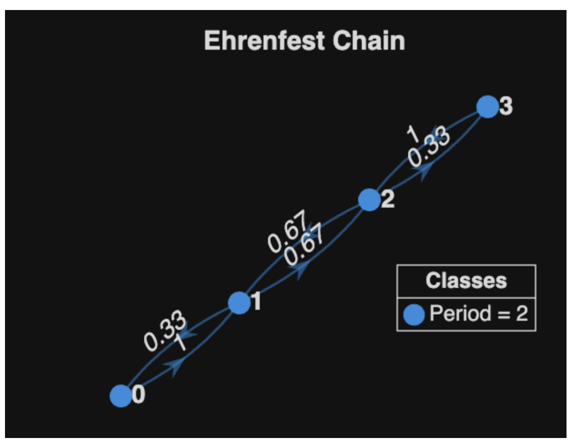

Figure 20.1: Ehrenfest Chain

Question 1. *If the matrix is doubly stochastic, then the uniform distribution $\pi(x)$ is uniform. Does it work the other way? I.E. if the stationary distribution is uniform, does that mean that the transition matrix is doubly stochastic.*

Solution 1. *Yes! Let $N = 3$, but it would work for any N value. This means we need to find the stopchastic matrix p such that*

$$\pi \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix} = \pi.$$

This means for one of the states, we would have $\pi = (1/3, 1/3, 1/3)$ *so* $\pi p = \pi$ *implies*

$$\sum_i \pi(i)p(i,j) = \frac{1}{3}\sum_i p(i,j) = \frac{1}{3}.$$

So all of the columns must add up to 1 for this to be the case $\implies$ *doubly stochastic.*

20.2 Return States

- We have used some notation to talk about probability or returning to states:
  - $\mathbb{P}_x(A) = \mathbb{P}(A \mid X_0 = x)$ means probability of an event A given we started at state x.
  - We let $T_y = \min\{n \geq 1 \mid X_n = y\}$ be the time of first return to y. This motivated:
    * $\rho_{yy} = \mathbb{P}_y(T_y < \infty)$ or "probability of returning to state y if you started at y.
    * $\rho(x, y) = \mathbb{P}_x(T_y < \infty)$ meaning it is probability of getting from x to y.
    * $\rho_{yy}^k = \mathbb{P}_y(T_y^k < \infty)$ or "probability of returning to state y k times if you started at y.
  - Now we will turn our attention to expected number of steps to return, or we will have to consider expected values for this probability and we will denote them by $\mathbb{E}_x$.
  - we also denote $N(y)$ be the number of visits to y at times $n \geq 1$.

Question 2. *What is the probability we go to* y k *times after starting at* x*?*

Proof: First, remember that the time of the k^{th} visit to y is defined as

$$T_y^k = \min\{n > T_y^{k-1} : X_n = y\}$$

and $\rho_{xy} = \mathbb{P}(T_y < \infty)$. Remembering that this is a memoryless problem (Strong Markov Property). First we need to go from state x to state y, which happens with probability ρ_{xy}. Then we need to go from y back to itself $k-1$ times, which is exactly ρ_{yy}^{k-1}. Putting this together, we have

$$\mathbb{P}_x(T_y^k < \infty) = \rho_{xy}\rho_{yy}^{k-1}. \tag{20.1}$$

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First we want to show that we can compute the expected value of X, which is $\mathbb{E}X$.

Lemma 20.2. *For any nonnegative integer valued random variable* X*, the expected value of* X *can be computed by*

$$\mathbb{E}X = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

Proof: We define $\mathbf{1}_{\{X \geq k\}}$ be a random variable that is 1 if $X \geq k$ and 0 otherwise. For example if we roll a fair die and get $X = 3$ then $\{X \geq k\} = \{1, 2, 3\}$. So $\mathbf{1}_{\{X \geq k\}} = \{1, 1, 1, 0, 0, 0, \cdots\}$. From this example, we can see that

$$X = \sum_{k=1}^{\infty} \mathbf{1}_{\{X \geq k\}}.$$

Now we take the expected value of both sides. But we notice that $\mathbb{E}\mathbf{1}_{\{X\geq k\}} = \mathbb{P}(X \geq k)$, so

$$\mathbb{E}X = \mathbb{E}\left[\sum_{k=1}^{\infty} \mathbf{1}_{\{X\geq k\}}\right] = \sum_{k=1}^{\infty} \mathbb{E}\mathbf{1}_{\{X\geq k\}} = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k)$$

■

Lemma 20.3. $\mathbb{E}_x N(y) = \frac{\rho_{xy}}{1-\rho_{yy}}$

Proof: Now the probability of returning at least k times, $\{N(y) \geq k\}$, is the same as the event that the k^{th} return occurs, i.e., $\{T_y^k < \infty\}$, so using Eqn 20.1, we have

$$\mathbb{E}_x N(y) = \sum_{k=1}^{\infty} \mathbb{P}\left(N(y) \geq k\right) = \rho_{xy} \sum_{k=1}^{\infty} \rho_{yy}^{k-1} = \frac{\rho_{xy}}{1-\rho_{yy}}.$$

The last equality happens because of the convergence of a geometric series, which we will show now. ■

Consider the geometric series:

$$S = \sum_{k=1}^{n} ar^n = a\left(1 + r + r^2 + r^3 + \cdots + r^n\right)$$

Multiply both sides by r:

$$rS = a\left(r + r^2 + r^3 + \cdots + r^{n+1}\right)$$

Now subtract the second equation from the first:

$$S - rS = a(1 + r + r^2 + \cdots + r^n) - a(r + r^2 + r^3 + \cdots + r^{n+1})$$

Most terms cancel:

$$S - rS = a\left(1 - r^{n+1}\right)$$

Factor the left-hand side:

$$S(1-r) = a\left(1 - r^{n+1}\right)$$

Solve for S:

$$S = \frac{a\left(1 - r^{n+1}\right)}{1-r}, \quad \text{for } r \neq 1$$

If $|r| < 1$, taking the limit as $n \to \infty$:

$$S = \frac{a}{1-r}$$

For our problem, we have $r = \rho_{yy}$, which has to be smaller than 1, so

$$\rho_{xy} \sum_{k=1}^{\infty} \rho_{yy}^{k-1} = \frac{\rho_{xy}}{1-\rho_{yy}}$$

Lemma 20.4. $\mathbb{E}_x N(y) = \sum_{n=1}^{\infty} p^n(x, y)$.

Proof: We start by reintroducing this notation $\mathbf{1}_{\{X_n=y\}}$ meaning that the variable is 1 only if $X_n = y$ and zero every other time. This means that

$$N(y) = \sum_{n=1}^{\infty} \mathbf{1}_{\{X_n=y\}}.$$

Taking the expected values gives

$$\mathbb{E}_x N(y) = \sum_{n=1}^{\infty} \mathbb{P}_x(X_n = y) = \sum_{n=1}^{\infty} p^n(x,y)$$

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Theorem 20.5. *y is recurrent if and only if*

$$\sum_{n=1}^{\infty} p^n(y,y) = \mathbb{E}_y N(y) = \infty.$$

Proof: The first equaility comes from whatr we just showed. The only difference is we start at y and not x. The second comes from the geometric series equation and $\rho_{yy} = 1$ if it is recurrent, so

$$\mathbb{E}_x N(y) = \frac{\rho_{xy}}{1-\rho_{yy}} = \frac{\rho_{xy}}{1-\rho_{yy}} = \frac{\rho_{xy}}{1-1} \to \infty$$

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Theorem 20.6. *Suppose that a Markov Chain is irreducible and all the states are recurrent. If $N_n(y)$ be the number of visits to y up to time n, then*

$$\frac{N_n(y)}{n} \to \frac{1}{\mathbb{E}_y T_y}$$

Theorem 20.7. *If the Markov chain is irreducible and there is a stationary distribution, then*

$$\pi(y) = \frac{1}{\mathbb{E}_y T_y}$$

Proof: Let X_0 has distribution π. From above, we see that

$$\frac{N_n(y)}{n} \to \frac{1}{\mathbb{E}_y T_y}.$$

Taking expected value and using the fact that $N_n(y) \leq n$, it can be shown that this implies

$$\frac{\mathbb{E}_\pi N_n(y)}{n} \to \frac{1}{\mathbb{E}_y T_y},$$

but since π is a stationary distribution $\mathbb{E}_\pi N_n(y) = n\pi(y)$. ■

	1	**2**	**G**	**D**
1	0.25	0.6	0	0.15
2	0	0.2	0.7	0.1
G	0	0	1	0
D	0	0	0	1

20.3 Exit Distributions

Example 20.8. *At a local two year college, 60% of fresh- men become sophomores, 25% remain freshmen, and 15% drop out. 70% of sophomores graduate and transfer to a four year college, 20% remain sophomores and 10% drop out. What fraction of new students eventually graduate?*

Solution 2. *We use a Markov chain with state space 1 = freshman, 2 = sophomore, G = graduate, D = dropout. The transition probability is*

Let $h(x)$ be the probability that a student currently in state x eventually graduates. By considering what happens on one step.

$$h(1) = 0.25h(1) + 0.6h(2)$$
$$h(2) = 0.2h(2) + 0.7$$

To solve we note that the second equation implies h(2) = 7/8 and then the first that

$$h(1) = \frac{0.6}{0.75} \cdot \frac{7}{8} = 0.7$$